Optimizer-free quantum simulation of sub-barrier neutron-pair transfer: resolution refinement and rodeo spectroscopy in the two-fragment pairing model

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The number-conserving two-fragment pairing model of Ref. [1] reduces sub-barrier neutron-pair transfer between superfluid nuclei to a time-dependent quasi-spin Hamiltonian that maps exactly onto longitudinal fields and number-conserving XY couplings on one qubit per pair slot. Running that model on a quantum device requires three things that the classical benchmark of Ref. [1] supplies for free: the correlated ground states of the two isolated fragments as the entrance state, a digital propagation through the collision pulse, and a readout that resolves the exit channel in pair number *and* in fragment excitation energy. We show how the first and the third of these can be done without any variational optimization by the two primitives that we implemented and assessed on the constant-pairing Hamiltonian in Ref. [2]: *resolution refinement*, the adiabatic transport of an eigenstate from a small to a larger active orbital space, and the *rodeo algorithm*, a projective spectral filter. Each fragment ground state is grown orbital by orbital from a single filled pair-slot configuration, so the entrance state of ^{116}Sn $+^{60}$ Ni (and the exit-channel reference states) are prepared with zero classical parameters; the same Trotterized XY evolution that implements the refinement path implements the collision; and a rodeo filter run with the two commuting fragment Hamiltonians after the pulse measures the exclusive, state-resolved transfer probabilities $P_{\kappa ab}$ from which the inclusive P_{2n} compared with experiment, the ground-state-to-ground-state channel, and a rigorous two-sided bracket $S_{\rm num} \le S_{T|P} \le S_{\rm diag}$ on the interfragment entanglement entropy follow. [Results sentence(s): refinement times, fidelities, gate budgets for 11, 19 and 30 qubits; comparison of the projector and the orbital-lowering refinement paths; rodeo resolution of the 0^+ strength in ^{42}Ca for the inclusive ^{96}Zr $+^{40}$ Ca data.] All stages are verified against the exact finite-space propagation of Ref. [1].

I. INTRODUCTION

[Paragraph 1: pairing, pair transfer, sub-barrier regime; one paragraph condensed from the companion draft.]

[Paragraph 2: why quantum simulation of transfer; the two-fragment pairing model as the compact, number-conserving, classically verifiable benchmark; 30 qubits for ^{116}Sn $+^{60}$ Ni, 26 for ^{96}Zr $+^{40}$ Ca; the model reproduces the measured D dependence with two calibrated parameters.]

Paragraph 3 (what is missing between the model and a device). The classical benchmark of Ref. [1] starts from the exact ground states of the isolated fragments, propagates every amplitude in the 5554-dimensional seniority-zero space, and reads off the complete transfer distribution and the reduced density matrix of the target. A device has none of this. It must (i) *prepare* the correlated entrance state $|\mathrm{GS}_T(2)\rangle \otimes |\mathrm{GS}_P(5)\rangle$ on 11 and 19 qubits; (ii) *propagate* it with a product formula through the Gaussian coupling pulse; and (iii) *read out* more than the pair-number partition, because the quantities that carry the physics — the ground-state-to-ground-state channel compared with exclusive data, the excited 0^+ strength lumped into inclusive data, and the entanglement entropy — live in the energy distribution within each transfer sector.

Paragraph 4 (what we bring from Ref. [2]). In Ref. [2] we implemented, on the constant-pairing Hamiltonian, a pipeline in which an eigenstate is transported adiabatically from a coarse to a fine single-particle basis along a shift-controlled, gapped path (resolution refinement [3]) and then cleaned by the rodeo algorithm [4, 5]. The assessment was sober: at eight qubits refinement did not beat a cheap UCCSD input on gate count; its merits were the absence of a classical optimization loop and of training measurements, and the fact that it is built from the same Trotterized evolution a device runs anyway. The two-fragment transfer model is a setting in which exactly those merits matter: the "coarse" states are *single computational basis states* (filled pair slots), the refinement, the collision and the rodeo controlled evolutions are all exponentials of one and the same two-local XY Hamiltonian, and the observable of interest is not a ground-state energy but a distribution over product eigenstates, which is precisely what a spectral filter measures.

Paragraph 5 (contributions and outline). [List: (a) orbital-by-orbital refinement chains for ^{60}Ni and ^{116}Sn from a filled configuration, two path variants (projector path of Ref. [2] and a two-local "orbital-lowering" path); (b) rodeo filtering of the refined fragment states; (c) the collision as the same Trotterized primitive, with an interaction-picture split that removes the large single-particle energies from the Trotter error; (d) two-fragment

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rodeo spectroscopy of the exit channel and the entropy bracket; (e) gate budgets per accepted preparation, collision and readout, verified against exact evolution. Outline of sections.]

II. THE TWO-FRAGMENT PAIRING MODEL

We summarize the model of Ref. [1] in the notation used there. Each fragment $X \in \{P, T\}$ is a set of active neutron orbitals p with single-particle energy ϵ_{pX} and pair degeneracy $\Omega_{pX} = (2j_{pX}+1)/2$. Restricted to seniority zero, the isolated fragment is the constant-strength pairing Hamiltonian

$$\hat{H}_X = \sum_{p\in X} 2\epsilon_{pX}\hat{n}_{pX} - G_X \sum_{p,p'\in X} \hat{P}^\dagger_{pX}\hat{P}_{p'X}, \qquad G_X = \frac{20.5}{A_X}\,\mathrm{MeV}, \tag{1}$$

with M_X active pairs, and the collision is

$$\hat{H}(t,D) = \hat{H}_P + \hat{H}_T + \hat{H}_{PT}(t,D), \quad \hat{H}_{PT} = -G_{PT}(t,D) \sum_{p\in P,\, q\in T} \left(\hat{P}^\dagger_{qT}\hat{P}_{pP} + \mathrm{h.c.}\right) \tag{2}$$

with a Gaussian pulse $G_{PT}(t,D) = G_{\mathrm{pk}}(D)e^{-(t/\tau(D))^2}$ and $\tau(D)$ fixed by the Coulomb curvature. For ^{116}Sn+^{60}Ni the active spaces are those of Table I of Ref. [1]: ^{60}Ni has $\Omega = (2,1,3,5)$ for $(2p_{3/2}, 2p_{1/2}, 1f_{5/2}, 1g_{9/2})$ with $M_T = 2$, and ^{116}Sn has $\Omega = (4,1,2,6,4,2)$ for $(1g_{7/2}, 3s_{1/2}, 2d_{3/2}, 1h_{11/2}, 2f_{7/2}, 3p_{3/2})$ with $M_P = 5$; the fixed-number dimensions are $d_T(2) = 9$, $d_P(5) = 154$, and the $M = 7$ collision space has dimension 5554.

a. Pair-slot encoding. Resolving every orbital into Ω_{pX} hard-core pair slots $i = (X, p, \lambda)$, with $\hat{P}^\dagger_{pX} = \sum_\lambda \hat{b}^\dagger_i$, is an exact representation of the seniority-zero space on the permutation-symmetric subspace of each orbital (Appendix B of Ref. [1]), and gives

$$\hat{H}(t,D) = E_c + \sum_i \zeta_i Z_i + \sum_{i<j} J_{ij}(t,D)\,(X_iX_j + Y_iY_j), \tag{3}$$

with $\zeta_i = -\epsilon_i + G_{ii}/2$ and $J_{ij} = -G_{ij}/2$, where $G_{ij} = G_X$ inside a fragment and $G_{PT}(t,D)$ across. For ^{116}Sn+^{60}Ni this is $N_q = 11 + 19 = 30$ qubits, seven of which are occupied, and the Hamiltonian has 30 single-qubit Z terms and $\binom{11}{2} + \binom{19}{2} + 11\cdot 19 = 55 + 171 + 209 = 435$ two-local XY terms, of which the 209 interfragment terms are switched on only during the pulse. Two features of Eq. (3) drive everything below. First, the encoding is *two-local*: no Jordan–Wigner strings appear, in contrast to the eight-qubit pairing model of Ref. [2] where a Trotter step of $k = 4$ levels required 325 Pauli strings of weight up to eight. Second, the total pair number $\hat{M} = \sum_i \hat{n}_i$ and each fragment's pair number $\hat{M}_X$ (outside the pulse) are conserved, so every stage stays in a fixed-M sector and the sector partition κ is read out by Z measurements on the target register alone.

[Fig. 1: pipeline schematic. Left: two registers (11 target slots, 19 projectile slots) grown orbital by orbital from filled configurations; rodeo ancilla. Middle: Gaussian pulse $G_{PT}(t)$, interfragment XY terms switched on. Right: Z readout of $\hat{M}_T$ plus two-fragment rodeo scan in (E_T, E_P).]

FIG. 1. Schematic of the optimizer-free pipeline. [caption]

b. Observables. After the pulse the state is analyzed in the product eigenbasis of $\hat{H}_P + \hat{H}_T$,

$$|\Psi(t_f,D)\rangle = \sum_\kappa \sum_{a,b} c_{\kappa ab}(D)\; |E^a_T(M_T{+}\kappa)\rangle \otimes \left|E^b_P(M_P{-}\kappa)\right\rangle, \tag{4}$$

where $a, b = 0$ label the fragment ground states. Ref. [1] compares P_{+1}/P_0 with $P_\kappa = \sum_{ab}|c_{\kappa ab}|^2$ to the measured $P_{2n}(D)$, and splits the target entanglement entropy as $S_{T|P} = S_{\mathrm{num}} + S_{\mathrm{cfg}}$ with $S_{\mathrm{num}} = -\sum_\kappa P_\kappa \log_2 P_\kappa$. At the reference distance $D = 13.4669$ fm the exact values are $P_0 = 0.97209$, $P_{+1} = 0.02206$, $P_{-1} = 0.00566$, $P_{+2} = 1.96\times10^{-4}$ and $S_{T|P} = 0.20622$ bits of which $S_{\mathrm{cfg}} = 4.31\times10^{-4}$ bits. These are the targets every stage below is checked against.

III. THE OPTIMIZER-FREE PIPELINE

Figure 1 summarizes the three stages: entrance-state preparation by refinement and rodeo (Secs. III A, III B), the Trotterized collision (Sec. III C), and exit-channel rodeo spectroscopy (Sec. III D). All three are exponentials of Eq. (3) with different coefficient schedules.

A. Resolution refinement of the fragment ground states

In Ref. [2] "resolution" meant the number k of doubly degenerate levels at fixed particle number. Here it means the set of active orbitals of a fragment at fixed pair number M_X, which is exactly the "larger active spaces" extension anticipated in the conclusions of Ref. [1], and, read backwards, the frozen-orbital approximation being

TABLE I. Candidate refinement chains for the ^{116}Sn $+^{60}$ Ni entrance channel. $|\mathcal{S}_r|$ is the number of pair slots (qubits), d_r the dimension of the seniority-zero (permutation-symmetric) space at fixed M_X, $\binom{|\mathcal{S}_r|}{M_X}$ the dimension of the fixed-number slot sector in which a gate-level simulation lives, and n_{XY} the number of XY terms of $\hat{H}_{\mathcal{A}_r}$. Overlaps p_r of the prolonged state $P_{r-1}|\mathrm{GS}_{r-1}\rangle$ with $|\mathrm{GS}_r\rangle$, minimum path gaps and extracted refinement times T_r^* are [to be computed].

X	$\mathcal{A}_r$	$\lvert\mathcal{S}_r\rvert$	d_r	$\binom{\lvert\mathcal{S}_r\rvert}{M_X}$	n_{XY}	p_r	T_r^*
T	$\{2p_{3/2}\}$	2	1	1	1	—	—
	$+\{2p_{1/2}, 1f_{5/2}\}$	6	5	15	15	[]	[]
	$+\{1g_{9/2}\}$	11	9	55	55	[]	[]
P	$\{1g_{7/2}, 3s_{1/2}\}$	5	1	1	10	—	—
	$+\{2d_{3/2}\}$	7	5	21	21	[]	[]
	$+\{1h_{11/2}\}$	13	26	1287	78	[]	[]
	$+\{2f_{7/2}, 3p_{3/2}\}$	19	154	11628	171	[]	[]

released adiabatically. Choose a chain of orbital sets

$$\mathcal{A}_0 \subset \mathcal{A}_1 \subset \cdots \subset \mathcal{A}_R = X, \tag{5}$$

and let $\mathcal{S}_r$ be the slots of $\mathcal{A}_r$. The prolongation P_r embeds a state on $\mathcal{S}_r$ into $\mathcal{S}_{r+1}$ by appending the new slots in $|0\rangle$; in the pair-slot encoding this is gate-free, as it was in canonical Jordan–Wigner order in Ref. [2], and the prolonged state satisfies $\langle P_r\psi_r|\hat{H}_{\mathcal{A}_{r+1}}|P_r\psi_r\rangle = \langle\psi_r|\hat{H}_{\mathcal{A}_r}|\psi_r\rangle$ exactly.

a. Chains. The occupations of Table I of Ref. [1] suggest the chains of Table I. Both start from a *filled* configuration, i.e. a single computational basis state: ^{60}Ni with two pairs in the two slots of $2p_{3/2}$, and ^{116}Sn with five pairs in the $4+1$ slots of $1g_{7/2}$ and $3s_{1/2}$. No variational or classical preprocessing of any kind is needed to start the chain, which is the analogue of the $k = 2 \to 4$ bootstrap of Ref. [2] where the $k = 2$ coarse state was the single filled determinant. [The exact overlaps of the filled states with the full ground states are to be computed; the mean occupations $\langle\hat{n}\rangle/\Omega$ of $3s_{1/2}$ (0.48) and $2d_{3/2}$ (0.29) in ^{116}Sn suggest that the first projectile step carries most of the correlation, whereas for ^{60}Ni the depletion of $2p_{3/2}$ (1.79 of 2 pairs) is mild.]

b. Path A: the projector path of Ref. [2]. Each step follows

$$H_r(s) = \cos^2\!\left(\tfrac{\pi s}{2}\right) P_r\big(\hat{H}_{\mathcal{A}_r} - \mu_r\big)P_r^\dagger + \sin^2\!\left(\tfrac{\pi s}{2}\right)\big(\hat{H}_{\mathcal{A}_{r+1}} - \mu_{r+1}\big), \tag{6}$$

with the shifts μ controlling the minimum gap as in Ref. [2]. Because the new slots are empty in the coarse space, the coarse term is the fine Hamiltonian projected onto the empty-new-slot subspace, $P_r\hat{H}_{\mathcal{A}_r}P_r^\dagger = \Pi_0\hat{H}_{\mathcal{A}_{r+1}}\Pi_0$ with $\Pi_0 = \prod_{i\in\mathcal{S}_{r+1}\setminus\mathcal{S}_r}|0_i\rangle\langle 0_i|$. This is the term whose circuit cost dominated the $3 \to 4$ chain in Ref. [2] (a multi-controlled phase on the new qubits, $\approx$ 350 CNOTs per application for two new qubits); here a step adds up to six new slots ($1h_{11/2}$), so the multi-controlled structure is correspondingly deeper and Path A is expected to be the expensive option. [Give the exact decomposition and CNOT count of $\Pi_0 H \Pi_0$ for the XY form: every XY term and Z term acquires the control string Π_0.]

c. Path B: orbital lowering (a two-local path). The same end points are connected by keeping the full two-local Hamiltonian and sweeping the single-particle energies of the new orbitals down from a large value,

$$H_r^\Lambda(s) = \hat{H}_{\mathcal{A}_{r+1}} + \Lambda\,\cos^2\!\left(\tfrac{\pi s}{2}\right)\sum_{i\in\mathcal{S}_{r+1}\setminus\mathcal{S}_r}\hat{n}_i, \tag{7}$$

i.e. $\epsilon_p(s) = \epsilon_p + \frac{1}{2}\Lambda\cos^2(\pi s/2)$ for the new orbitals. At $s = 0$ and $\Lambda \gg G_X\Omega$ the ground state of Eq. (7) is the prolonged coarse state up to corrections of order $(G_X\Omega/\Lambda)^2$ (second-order perturbation theory in the pair-scattering into the penalized slots eliminates them and reproduces $\Pi_0\hat{H}\Pi_0$ to leading order), so Path B is a two-local relaxation of Path A. Along Path B every Trotter step costs exactly what a collision step costs, and the path has a direct physical reading: the frozen orbitals of the fragment are lowered adiabatically from infinity to their physical position. [Compute gap, overlap and T^* along both paths for every chain step; the comparison of $T^*\times$(cost per step) between A and B is one of the new results of this paper. Check the Λ dependence of the initial infidelity and of T^*; a schedule with $\Lambda \to \infty$ at $s = 0$, e.g. $\Lambda\cot^2(\pi s/2)$ regularized, may be preferable.]

d. Refinement time and sufficient conditions. We extract T^* as in Ref. [2]: the smallest T at which the overlap of the transported state with the exact fine ground state has reached a stated target, with the $\sin^2$ schedule and the Jansen–Ruskai–Seiler bound [6] as the reference whose over-estimation we again quantify. New here is that the fragment Hamiltonians are Richardson-type with realistic, non-uniform level spacings and degeneracies $\Omega_p > 1$; the ratios $G_X/\Delta\epsilon$ are $\approx$ 0.1–0.3 ($G_{^{116}\mathrm{Sn}} = 0.177$ MeV, $G_{^{60}\mathrm{Ni}} = 0.342$ MeV), i.e. the fragments are on the weak-coupling side of the $g = 1$, $\delta = 1$ model of Ref. [2], and the minimum path gaps are set by the lowest seniority-zero excitation of each fragment, which in turn is set by the orbital spacings rather than by the pairing. Times are in units of $\hbar/\mathrm{MeV} = 6.58 \times 10^{-22}$ s; the collision pulse lasts $10\tau = 3.42\,\hbar/\mathrm{MeV}$ for ^{116}Sn $+^{60}$ Ni, which is the natural yardstick for T^*. [Table or figure of spectra along each path; discuss whether the orbital-spacing-dominated gaps make refinement *faster* than in the $g = 1$ model.]

B. Rodeo filtering of the refined fragment states

Each refined fragment state is cleaned by the rodeo algorithm with $E_{\mathrm{obj}} = E_X^{(0)}(M_X)$, exactly as in Sec. III.D of Ref. [2]: M cycles of a Hadamard, the controlled evolution $e^{-i\hat{H}_X t_k}$, the phase $e^{iE_{\mathrm{obj}}t_k}$ and a measurement of

a single reused ancilla; the ensemble infidelity obeys

$$1-\mathcal{F}_M = \frac{\sum_{k\neq 0}|c_k|^2 q_k^M}{p+\sum_{k\neq 0}|c_k|^2 q_k^M}, \qquad q_k = \tfrac{1}{2}\big(1+e^{-(E_k-E_0)^2\sigma^2/2}\big). \tag{8}$$

Three things differ from the first paper. (i) The two fragments are uncoupled at t_i, so the entrance state is prepared as two independent problems on 11 and 19 qubits, each with its own $E_{\rm obj}$, and the ancilla can be shared. (ii) $E_{\rm obj}$ is known classically for the benchmark ($d_P(5) = 154$), but a rodeo *scan* on the refined state locates it on a device as well and doubles as a verification that the refinement reached the right state — the energy-scan mode of the algorithm that Ref. [2] used only for illustration. (iii) The controlled evolution is two-local: a controlled XY rotation costs a fixed small number of CNOTs, and the controlled-reversal-gate construction of Ref. [7] applies directly to Z and XY terms. The width σ is chosen from the fragment spectrum so that $\Delta_{\min}\sigma \gtrsim 5$ as before; [quote $\Delta_{\min}$ of each fragment at fixed M_X once computed].

[Decide whether to keep a variational comparator. The natural one in the XY encoding is pair-UCCD (pUCCD), a product of XY rotations acting on the filled configuration; it is cheap and would play the role UCCSD played in Ref. [2]. Given the main finding there, the honest framing is: report the matched-observable comparison, but the selling point of the present pipeline is that no optimizer is needed anywhere between the filled configuration and the final readout.]

C. The collision as the same Trotterized primitive

The pulse is propagated with a second-order product formula in the midpoint rule for the time dependence of G_{PT},

$$U(t_f,t_i) \approx \prod_n e^{-i\hat{H}_Z\delta t/2}\, e^{-i\hat{H}_{XY}(\bar{t}_n)\delta t}\, e^{-i\hat{H}_Z\delta t/2}, \tag{9}$$

where $\hat{H}_Z$ collects the 30 longitudinal terms (which commute and are exponentiated exactly by single-qubit rotations) and $\hat{H}_{XY}(t)$ the 435 exchange terms, grouped into layers of commuting pairs. Two observations specific to this model.

a. Interaction picture with respect to $\hat{H}_Z$. The single-particle energies are ~ 10 MeV while the couplings are $\lesssim 0.35$ MeV, so the Trotter error of Eq. (9) is dominated by $[\hat{H}_Z, \hat{H}_{XY}]$. Since $\hat{H}_Z$ is a sum of commuting single-qubit terms, the rotating frame $\tilde{H}_{XY}(t) = e^{i\hat{H}_Z t}\hat{H}_{XY}(t)e^{-i\hat{H}_Z t}$ is still two-local — each $X_iX_j+Y_iY_j$ becomes $\cos(\Delta_{ij}t)(X_iX_j+Y_iY_j)+\sin(\Delta_{ij}t)(X_iY_j-Y_iX_j)$ with $\Delta_{ij} = 2(\zeta_i-\zeta_j)$ — and its Trotter error is governed by G^2 rather than $G\epsilon$. Over the pulse duration $3.4\,\hbar$/MeV and with $G_{\rm pk} \leq 0.3$ MeV this suggests that a modest number of steps suffices; [quantify: $P_\kappa(D)$ and $S_{T|P}$ versus δt in both frames against the DOP853 reference of Ref. [1]; report the step count at which the three-order-of-magnitude fall of $P_{2n}(D)$ in Fig. 2 of that paper is reproduced to within the experimental scatter]. The same observation applies to the refinement paths, whose Trotterization in Ref. [2] was found to be far more forgiving than the sufficient conditions suggest.

b. Trotter leakage out of the symmetric subspace. The exact evolution preserves the permutation-symmetric subspace of every orbital, but an individual XY term between two slots of the same orbital does not, so a product formula leaks weight at $O(\delta t^2)$ into configurations that are not seniority-zero states of the model. These configurations have different pairing energies (the diagonal term $G_X n(\Omega-n+1)$ is specific to the symmetric irrep), so the leakage is energetically resolvable: it shows up as extra weight in a rodeo scan and is removed by the exit-channel filter. [Measure the leaked weight versus δt; it is a cheap and specific Trotter diagnostic for this encoding. Consider ordering the within-orbital terms so that the collective exponential $e^{-i\theta\hat{P}_p^\dagger\hat{P}_p}$ is applied as one block.]

c. Pulse truncation. Ref. [1] integrates over $\pm 5\tau$ where the coupling is e^{-25} of its peak. On a device one truncates to $\pm 3\tau$ ($e^{-9} \approx 10^{-4}$) or less; [report the effect on P_κ].

d. An adiabatic bound on the non-elastic probability. The pulse returns $\hat{H}(t,D)$ to $\hat{H}_P+\hat{H}_T$, so the entrance state is the instantaneous ground state at both ends and $1-P_{000}$, the probability of having left it, is bounded by the adiabatic theorem with the pulse as the schedule — the same Jansen–Ruskai–Seiler expression we evaluated for the refinement path in Sec. IV.E of Ref. [2]. Since those bounds overestimated the actual leakage by two to three orders of magnitude, we expect the bound on $1-P_{000} \approx 0.03$ to be loose; [evaluate it anyway, as a link between the two papers and as a statement about how non-adiabatic sub-barrier transfer is in this model].

D. Rodeo spectroscopy of the exit channel

This is the stage where the rodeo algorithm does something the first paper did not ask of it. After the pulse $\hat{H} = \hat{H}_P+\hat{H}_T$ with $[\hat{H}_P,\hat{H}_T]=0$, so the two fragment Hamiltonians can be filtered *jointly*: M_T rodeo cycles with $e^{-i\hat{H}_T t_k}$ and energy E_T, followed by M_P cycles with $e^{-i\hat{H}_P t'_k}$ and energy E_P, with one shared ancilla or two. For the state of Eq. (4) the all-zero acceptance probability averaged over Gaussian time sets is

$$P_{0^M}(E_T,E_P) = \sum_{\kappa ab}|c_{\kappa ab}|^2\, q^{M_T}\big(E_T^a(M_T+\kappa)-E_T\big)\, q^{M_P}\big(E_P^b(M_P-\kappa)-\tag{10}$$

with $q(\Delta) = \tfrac{1}{2}(1+e^{-\Delta^2\sigma^2/2})$. A scan in the (E_T,E_P) plane therefore displays the final state as peaks at the product energies, with heights that converge geometrically to the exclusive probabilities $|c_{\kappa ab}|^2$. Reading out

$\hat{M}_T$ on the target register in the same shot assigns each peak its sector κ and removes accidental degeneracies between products belonging to different sectors. Three uses follow.

a. Exclusive versus inclusive transfer. The quantity compared with the ^{116}Sn $+^{60}$ Ni data in Ref. [1] is P_{+1}/P_0, the full one-pair-pickup sector. The filter with $E_T = E_T^{(0)}(3)$, $E_P = E_P^{(0)}(4)$ measures the ground-state-to-ground-state channel $|c_{+1,00}|^2$ alone, and the residual $P_{+1} - |c_{+1,00}|^2$ is the excited 0^+ strength of ^{62}Ni$\otimes^{114}$Sn populated in the model. For ^{96}Zr $+^{40}$ Ca the situation is reversed: the data are inclusive and Ref. [1] had to place the centroid of the +1 sector at $Q = 0$ by hand; the two-fragment scan resolves the 0^+ strength in ^{42}Ca orbital by orbital and makes the inclusive-channel approximation explicit. Note that the entrance space of ^{40}Ca is empty, so for this reaction only ^{96}Zr (three pairs in five orbitals, [slot count]) needs to be prepared, and the target register starts in $|0\cdots0\rangle$: ^{96}Zr $+^{40}$ Ca is the simplest complete end-to-end case and the natural first calculation.

b. Measuring a small-weight channel. The filter does not need to *prepare* the transferred state with high fidelity to *measure* its weight: the acceptance probability obeys $P_M = p + \sum_{k\neq 0} |c_k|^2 q_k^M$ and p is read off the plateau in M. For $p = |c_{+1,00}|^2 \approx 0.02$ the background $2^{-M}(1-p)$ falls below p at $M \approx 6$ and below $10^{-2}p$ at $M \approx 13$; for the two-pair channel $P_{+2} \approx 2\times10^{-4}$ one needs $M \approx 19$. These cycle counts, times the controlled-evolution cost of Sec. III B, set the readout budget; the acceptance rate is of course $\approx p$ itself, so the shot overhead is $1/p$, as for any measurement of a rare channel. [Table of M, σ, acceptance and CNOT count per accepted shot for $\kappa = 0, \pm1, +2$.]

c. A two-sided bracket on the entanglement entropy. Because $[\rho_T, \hat{M}_T] = 0$, Ref. [1] writes $S_{T|P} = S_{\rm num} + S_{\rm cfg}$ with $S_{\rm cfg} \geq 0$, so $S_{\rm num} \leq S_{T|P}$; $S_{\rm num}$ needs only the Z readout of $\hat{M}_T$. On the other side, the Shannon entropy of the diagonal of ρ_T in any orthonormal basis is an upper bound on its von Neumann entropy, and the rodeo scan with $\hat{H}_T$ alone measures exactly that diagonal in the eigenbasis of $\hat{H}_T$, $P_{\kappa a} = \sum_b |c_{\kappa ab}|^2 = \langle E_T^a|\rho_T|E_T^a\rangle$. Hence

$$S_{\rm num} = -\sum_\kappa P_\kappa \log_2 P_\kappa \;\leq\; S_{T|P} \;\leq\; S_{\rm diag} = -\sum_{\kappa,a} P_{\kappa a}\log_2 P_{\kappa a}, \tag{11}$$

and both ends are measured by the same circuit. Since Ref. [1] finds $S_{\rm cfg}/S_{T|P} < 0.3\%$ at the reference distance and $\leq 2\%$ at the smallest, the bracket is expected to be tight across the whole D range; where it is not, its width is itself a measure of the coherence within a transfer sector that the diagonal distribution misses. [Compute $S_{\rm diag}(D)$ from the exact final states and plot the bracket against $S_{T|P}(D)$ for both reactions. The bracket is, to our knowledge, the first proposal for a device-level estimate of the interfragment entanglement of Ref. [1]; check the literature on diagonal/"observational" entropy bounds and cite.]

E. Optional: the static di-nuclear ground state

[Freeze the coupling at its peak value $G_{\rm pk}(D)$ and ask for the ground state of the coupled 30-qubit Hamiltonian. The refinement path $H(s) = \hat{H}_P + \hat{H}_T + s\,G_{\rm pk}\hat{V}_{PT}$ connects the product entrance state to it ("coupling refinement", $P = \nVdash$, two-local throughout), and the rodeo filter finishes it. Its interfragment entropy and pair-number distribution are the static counterparts of the dynamical ones and give an adiabatic reference for the sub-barrier transfer. Decide whether this belongs in this paper or in a follow-up; it is cheap once the pipeline exists.]

IV. WHAT CARRIES OVER FROM THE PAIRING-MODEL STUDY

[Short prose around Table II. Emphasize the two lessons of the first paper that shaped the design here: (1) the coarse projector term is the cost driver, hence Path B; (2) refinement earns its place where it replaces an optimizer rather than where it saves gates, hence the filled-configuration bootstrap.]

V. RESULTS

[To be written from the notebook. Planned subsections and figures:]

1. **Fragment spectra and path gaps.** Low-lying seniority-zero spectra of ^{60}Ni and ^{116}Sn at each chain step; minimum gaps along Paths A and B versus μ and Λ. (Fig. 2)

2. **Refinement: overlap and energy versus** T for every chain step and both paths; T^*; comparison with the single-shot $\mathcal{A}_0 \to \mathcal{A}_R$ refinement. (Fig. 3, Table I)

3. **Rodeo filtering of the fragments:** $1 - \mathcal{F}_M$ versus M for refined inputs at T^* and at a longer T; energy scans as a self-check. (Fig. 4)

4. **Trotterized collision:** $P_\kappa(D)$ and $S_{T|P}(D)$ versus step size, laboratory versus interaction picture, symmetric-subspace leakage, pulse truncation; overlay on Fig. 2 of Ref. [1]. (Fig. 5)

5. **Exit-channel spectroscopy:** the (E_T, E_P) scan at $D = 13.4669$ fm; exclusive g.s.-to-g.s. probability versus D against the inclusive P_{+1}/P_0; the ^{42}Ca 0^+ strength for ^{96}Zr $+^{40}$ Ca. (Fig. 6)

6. **Entropy bracket** $S_{\rm num} \leq S_{T|P} \leq S_{\rm diag}$ versus D for both reactions. (Fig. 7)

TABLE II. Ingredients of Ref. [2] and their status in the present work.

Ingredient	In Ref. [2]	Here
Model	constant pairing, k equispaced doubly degenerate levels, $N = 4$, $g = 1$; optional particle-hole term	two Richardson-type fragments with realistic ϵ_p, $\Omega_p \leq 6$, $M_P = 5$, $M_T = 2$; time-dependent interfragment coupling
Encoding	Jordan–Wigner, $2k$ qubits, Pauli strings of weight up to $2k$	pair slots, one qubit per slot, strictly two-local Z/XY
Coarse state	UCCSD–VQE or the filled $k = 2$ determinant	filled slot configuration (no parameters)
Prolongation	gate-free in JW order	gate-free
Refinement path	shift-controlled projector path, Eq. (6)	Path A unchanged; Path B (orbital lowering) new
T^* protocol, JRS bound	Sec. III.C, IV.E	reused unchanged
Rodeo filter	ground-state preparation, Eqs. (8)	reused for fragments; new use as two-fragment spectroscopy, Eq. (10)
Comparators	prolonged, UCCSD, refined inputs to the same filter	[pUCCD?]; main comparison is Path A vs Path B and prep/collision/readout budgets
Cost accounting	CNOTs per accepted preparation, coarse term included	same, plus collision and readout per accepted shot
Verification	Richardson exact solution, exact sector evolution	exact fragment diagonalization and DOP853 propagation of Ref. [1]
Code	`pairinglib`, notebook, 26 tests	[generalize to $\Omega_p > 1$, non-uniform ϵ_p, two registers, time-dependent J_{ij}; fixed-M slot-sector simulator ($\binom{30}{7} = 2.0 \times 10^6$ amplitudes for the full collision)]

7. **Gate budgets:** CNOTs for preparation (per fragment, per path), collision (per Trotter step and total) and readout (per accepted shot, per channel); total per accepted exclusive event on 30 qubits; verification of every stage against exact evolution. (Fig. 8, Table III)

8. **Sufficient conditions versus practice** for the refinement paths and the adiabatic bound on $1 - P_{000}$ for the pulse.

VI. DISCUSSION AND OUTLOOK

[Points to make: the pipeline is a single Trotterized XY evolution with three coefficient schedules and one ancilla; what the two-local encoding buys relative to the JW pairing model; the price of 30 qubits and of the rare-channel shot overhead; where refinement could lose (long T^* if the first projectile step has a low overlap or a small gap — here a longer chain or a pUCCD warm start are the fallbacks, and "optimized refinement" in the sense of the nucleation paper [8] is the variational one); extensions listed in Ref. [1] — larger spaces, broken pairs, orbital-dependent couplings — and which of them the present primitives survive (broken pairs: yes, cf. the particle-hole section of Ref. [2], at the price of leaving the two-local pair-slot encoding); hardware: the 11-qubit ^{60}Ni preparation and the 26-qubit ^{96}Zr + ^{40} Ca collision as device targets.]

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CODE AVAILABILITY

[Repository, notebook, tests.]

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